Linear Algebra I: HW9 Partial Solution

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1. Let X_{ij} be the $n \times n$ matrix with the $(i,j)-$entry being 1 and 0 otherwise, given $A = (A_{ij}) \in M_{n\times n}(F)$, compute that $X_{ij}A$ is the matrix with i-th row being $(A_{j1}, \cdots, A_{jn})$ and others being 0, and AX_{ij} is the matrix with the j-th column being $(A_{1i}, \cdots, A_{ni})^t$ and others being 0. The rest of the proof follows from this observation.

3-1-5. It suffices to prove that if E is an elementary matrix, then E^t is an elementary matrix (since $(E^t)^t = E$). Let X_{ij} be the $n \times n$ matrix with the $(i,j)-$entry being 1 and 0 otherwise. Recall that there are three types of elementary matrices:

Type 1: Interchanging i,j-th row (column)

$$E_{ij} := I_n - X_{ii} - X_{jj} + X_{ij} + X_{ji}$$

Type 2: Multiplying i-th row (column) by a nonzero scalar a

$$D_{i,a} := I_n - X_{ii} + aX_{ii}$$

Type 3: Adding $a\times$ i-th row to j-th row (or adding $a\times$ j-th column to i-th column)

$$A_{ij,a} = I_n + aX_{ji}$$

Then $E_{ij}^t = E_{ij}, D_{i,a}^t = D_{i,a}, A_{ij,a}^t = A_{ji,a}$, thus the transpose of any elementary matrix is still elementary.

3-1-6. If B can be obtained from A by elementary column operations, then $B = AE_1E_2\cdots E_n$. Thus,

$$B^t = (AE_1E_2\cdots E_n)^t = E_n^t\cdots E_2^tE_1^tA^t.$$

Since the transpose of elementary matrices are elementary, B^t can be obtained from A^t by elementary row operations. The proof from row operation is almost identical.

3-1-8. If Q can be obtained from P by an elementary row operation, then $Q = EP$ for some E. We have three cases:

E is type 1: $E = E_{ij} := I_n - X_{ii} - X_{jj} + X_{ij} + X_{ji}$ for some i,j, then $E^{-1} = E_{ij} = E$ is also type 1.

E is type 2: $E = D_{i,a} := I_n - X_{ii} + aX_{ii}$ for some i and $a \neq 0$, then $E^{-1} = D_{i,a^{-1}}$ is also type 2.

E is type 3: $E = A_{ij,a} = I_n + aX_{ji}$ for some i,j and $a \in F$, then $E^{-1} = A_{ij,-a}$ is also type 3.

Therefore, $E^{-1}Q = P$, that is, P can be obtained from Q by an elementary row operation of the same type.

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3-2-11. We claim that $\operatorname{im} L_B = \{(a, z) \mid a \in F, z \in \operatorname{im} L_{B'}\}$, if $v \in RHS$, $v = (a, Bw)^t$ for some $a \in F, w \in F^n$, define $v' = (a, w)^t \in F^{n+1}$, then

$$Bv' = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & B' & \\ 0 & & & \end{array}\right) \begin{pmatrix} a \\ \\ w \\ \\ \end{pmatrix} = v.$$

Therefore, $RHS \subseteq LHS$. Similarly, if $x = (b, y)^t \in \operatorname{im} L_B$ where $y \in F^m$, then $y \in \operatorname{im} L_{B'}$, and the proof of claim is done. Note that $\operatorname{im} L_B \cong F \oplus \operatorname{im} L_{B'}$, thus $n = \operatorname{rank} L_B = \operatorname{rank} L_{B'} + 1$, and $\operatorname{rank} B' = \operatorname{rank} L_{B'} = n - 1$.